Yejjipurapu Ganesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.60
Jawahar Navodaya vidyalaya,guntur	Class XII (CBSE)	2023	94.80%
Jawahar Navodaya vidyalaya, guntur	Class X (CBSE)	2021	76.00%

SKILLS

Languages : C,C++, Python, SQL, MATLAB, HTML/CSS,JavaScript

Frameworks: Numpy, Pandas, Matplotlib, Seaborn

Tools: Git/Github

PROJECTS

Full-Stack Authentication System | *Python, FastAPI, JavaScript, JWT, Render, Git* |*GitHub*

- **Engineered** a RESTful backend API using Python and FastAPI, handling user registration, authentication, and secure token blacklisting.
- **Implemented** industry-standard security by hashing passwords with bcrypt and managing stateless user sessions via JSON Web Tokens (JWT).
- **Architected** Role-Based Access Control (RBAC), restricting sensitive endpoints and administrative dashboards exclusively to verified admin users.
- **Developed** a responsive frontend client using JavaScript and HTML/CSS, utilizing the asynchronous fetch API for seamless client-server communication.
- **Deployed** the complete system to the cloud, hosting the backend on Render and the frontend on GitHub Pages to establish a live, production-ready environment.

File Compression Tool | Github

- Implemented a **Huffman Coding-based file compression utility** in C++ for lossless text compression and decompression.
- Optimized encoding using **greedy algorithms, binary trees, and STL containers**, achieving efficient memory and runtime performance.
- Developed a modular CLI application featuring file parsing, binary serialization, and validation of decompressed output.

Four-bar and slider-crank simulator

- Implemented a **Python simulator for four-bar and slider-crank mechanisms**, generating kinematic plots and animations to study link motion and transmission characteristics.
- Automated kinematic analysis by calculating **position, velocity, and acceleration profiles**, reducing manual computation and improving design evaluation.

POSITIONS OF RESPONSIBILITY

Head of Multimedia | **Milan**

- Leading the Multimedia Team by planning and overseeing the design of digital content, event branding, and promotional campaigns.
- Managing multimedia operations and collaborating with organizing teams to deliver high-quality visual assets within deadlines.

Coordinator at **Canteen Monitoring Committee** (CMC - IITH)

- Evaluated food quality and hygiene practices, documented student concerns, and collaborated with vendors and

institute authorities to address operational issues.

PR Coordinator, **Diesta**

- Coordinated public relations activities, managed communication with participants and external stakeholders, and promoted the event to maximize outreach and engagement.

RELEVANT COURSES

Mathematics : Calculus, Linear Algebra, Differential Equations, Introduction to Probability, Introduction to Statistics, Transform Techniques, Complex Numbers.

Computer Science : Introduction to Programming.

Others : Introduction to Enterpreneurship, Communication Skills

EXTRACURRICULAR

National Sports Organisation (NSO): Active participant in institute sports and physical fitness activities.

Milan 2025 (Inter-Hostel Sports Fest): Participated in **Athletics**, demonstrating discipline, perseverance, and teamwork.